

Heughric Fernando | Section: CM3 | CSS154 | Unix Environment Setup

Screenshot 1:

```
heugh@Azyuri-Bravo15:/mnt/c/School/3rdYear/3rd Ter,/CSS154/CSS154/UnixEnv$ docker run --rm -it -v "$(pwd)":/workspace -w /workspace nvidia/cuda:12.4.0-devel-ubuntu20.04 /bin/bash

== CUDA ==

CUDA Version 12.4.0

Container image Copyright (c) 2016-2023, NVIDIA CORPORATION & AFFILIATES. All rights reserved.

This container image and its contents are governed by the NVIDIA Deep Learning Container license.
By pulling and using the container, you accept the terms and conditions of this license:
https://developer.nvidia.com/ngc/nvidia-deep-learning-container-license

A copy of this license is made available in this container at /NGC-DL-CONTAINER-LICENSE for your convenience.

WARNING: The NVIDIA Driver was not detected. GPU functionality will not be available.
Use the NVIDIA Container Toolkit to start this container with GPU support; see
https://docs.nvidia.com/datacenter/cloud-native/.

root@ea2508f38288:/workspace#
```

Screenshot 2:

```
root@ea2508f38288:/workspace/unixenv_cuda# nvcc -Xcompiler -fPIC -shared -o libmylib.so mylib.cu
root@ea2508f38288:/workspace/unixenv_cuda# g++ -o main main.cpp -I. -L. -lmylib -Wl,-rpath,. -ldl
root@ea2508f38288:/workspace/unixenv_cuda# nvcc -Xcompiler -fPIC -shared -o libmylib2d.so mylib2d.cu
root@ea2508f38288:/workspace/unixenv_cuda# g++ -o main2d main2d.cpp -I. -L. -lmylib2d -Wl,-rpath,. -ldl
root@ea2508f38288:/workspace/unixenv_cuda# ls -lh libmylib.so libmylib2d.so main main2d
-rwxrwxrwx 1 1000 1000 978K Jun  2 21:43 libmylib.so
-rwxrwxrwx 1 1000 1000 978K Jun  2 21:43 libmylib2d.so
-rwxrwxrwx 1 1000 1000  18K Jun  2 21:43 main
-rwxrwxrwx 1 1000 1000  18K Jun  2 21:43 main2d

!./main
[8] ✓ 0.7s Python
... 11 22 33 44 55
```

Screenshot 3:

```
!./main2d
[13] ✓ 0.5s Python

Input Array A:
1    2    3    4
5    6    7    8
9   10   11   12

Input Array B:
10   20   30   40
50   60   70   80
90   100  110  120

Output Array C (A + B):
11   22   33   44
55   66   77   88
99   110  121  132
```